SCOPE : L3 — Self-Consistency

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Chapter 0: Introduction

The Sephirotic names used in this paper are not intended in a religious or mystical sense. In SCOPE, these names are borrowed as labels to identify cognitive structures.

SCOPE (Sephirotic Cognitive Process Engine) is a coordinate system for estimating the cognitive structures of humans and AI. Here, "coordinate system" does not mean a flowchart in which cognitive processing proceeds in a fixed order. It is a framework for organizing and estimating cognitive structure. Each layer does not indicate hierarchical rank or temporal sequence.

In SCOPE, the state in which the result of cognitive processing is output into reality and becomes observable is positioned as "Malkuth." What can be observed is only the output itself; it is not possible to directly know what cognitive structure operated internally to produce it. For this reason, SCOPE uses observable output as a clue to estimate the cognitive structure behind it.

In reality, there are cases where a person's position does not change even when sufficient counter-evidence is presented. There are also cases where a person appears to hold seemingly contradictory values or beliefs at the same time. These cases may not be fully explained by logical ability or knowledge alone.

This paper addresses one such cognitive structure: L3. L3 is the cognitive layer responsible for self-consistency within the individual. The definition and characteristics of L3 are discussed in detail in Chapter 1.

Note that L3 is not defined as the starting point of cognitive processing. SCOPE does not assume that cognition begins from any specific layer; rather, it treats each layer as part of a cognitive structure in which layers mutually influence one another.

This paper examines the characteristics and basic properties of L3 as a self-consistency layer, as well as its functional correspondence in humans and AI. Some points remain at a hypothesis stage and are left as topics for future consideration.

Chapter 1: What Is L3

L3 is the cognitive layer responsible for self-consistency.

Self-consistency, as used here, refers to the process by which an individual organizes values, beliefs, worldview, and self-image internally into a form with minimal contradiction. In SCOPE, L3 is positioned as the cognitive layer used to organize and estimate this process.

L3 is not a cognitive layer that guarantees agreement with reality or logical correctness. Nor is it a layer that judges truth or falsity. Self-consistency may hold within the individual even when it appears to contradict reality.

L3, as used here, does not refer to a biological organ or a specific region of the brain. In SCOPE, it is positioned as a cognitive layer for estimating how self-consistency is organized within the individual.

This paper positions L3 as the self-consistency layer and organizes its characteristics in the following chapters.

Chapter 2: Basic Characteristics of L3

Chapter 1 defined L3 as the cognitive layer responsible for self-consistency. Building on that definition, this chapter organizes the basic characteristics of L3.

L3 is the cognitive layer that maintains self-consistency within the individual. Self-consistency, as used here, refers to the process of organizing values, beliefs, worldview, and self-image into a form with minimal internal contradiction. This self-consistency does not, however, guarantee agreement with reality or logical correctness.

L3 is also not a layer that holds only a single self-consistency at all times. Depending on the situation, multiple self-consistencies may hold at the same time.

Furthermore, the self-consistencies handled by L3 are not uniform. They range from everyday preferences to values, beliefs, worldview, and self-image. Each of these differs in nature.

Self-consistency is also not always stable. When observed facts or experiences increase the contradiction with existing self-consistency, self-consistency may change. However, the mechanism of this change requires consideration of its relationship with other layers, and is not addressed in this paper.

This paper positions L3 as a cognitive layer with these basic characteristics. The next paper examines how this change in self-consistency may occur.

Chapter 3: L3 Changes

Chapter 2 organized L3 as a cognitive layer that maintains self-consistency, and noted that self-consistency may change under certain conditions. This chapter examines the conditions under which such change may occur.

L3's basic role is to maintain self-consistency. Self-consistency is maintained as far as possible, but when maintaining it becomes difficult, it may change. This change is not a loss of self-consistency, but something that occurs in order to preserve it.

The degree of change in self-consistency does not necessarily correspond to the scale of the triggering event. Self-consistency may change in response to a minor event, while in other cases it may be maintained even when sufficient counter-evidence is presented. This difference may also depend on the nature of the self-consistency in question, but this paper does not go further into that point.

What matters here is that this change should not be treated simply as an increase or decrease in knowledge. Gaining new knowledge or experience is not the same as a change in self-consistency.

The process by which self-consistency changes is not addressed in this paper. This change requires consideration of its relationship with other layers.

As described above, L3 is positioned as a cognitive layer that fundamentally maintains self-consistency, while allowing for change under certain conditions. The next paper examines how L3 may relate to other layers.

Chapter 4: L3 Interacts with Other Layers

Chapter 3 organized L3 as a cognitive layer that fundamentally maintains self-consistency, while allowing for change under certain conditions. This chapter examines L3's relationship with other layers.

L3 does not function in isolation. Self-consistency within the individual may be influenced by a variety of cognitive processes. For this reason, L3 is positioned as a cognitive layer that functions in relation to other layers.

At the same time, the purpose of this paper is to organize the role of L3 on its own. For that reason, the specific processes by which each layer contributes, and how they influence self-consistency, are not addressed in this paper.

In SCOPE, each layer is defined as an independent cognitive function, while also being treated as part of a cognitive structure in which layers mutually influence one another. L3 is no exception. Self-consistency does not arise independently of other layers, but functions through interaction with them.

As described above, L3 does not function in isolation, but operates in relation to other layers. The specific interactions between L3 and other layers will be examined separately within the SCOPE framework as a whole.

Chapter 5: L3 and Its Relationship to Malkuth

Chapter 4 organized L3 as a cognitive layer that functions in relation to other layers. This chapter examines how L3 can be observed through its relationship with Malkuth.

Self-consistency is an internal cognitive structure, and its state cannot be directly observed. In SCOPE, cognition itself cannot be directly observed. What can be observed is Malkuth, the state in which the result of cognitive processing appears in reality. For this reason, self-consistency in L3 also cannot be directly observed.

For example, the same statement, "I'm fine," may reflect a case where there is truly no problem, or a case where a person is telling themselves "I'm fine" as a form of reassurance. Even when the observed statement is identical, the self-consistency underlying it is not necessarily the same. For this reason, L3 cannot be determined from a single observed result alone.

This point is not unique to L3; it applies to other layers as well. What SCOPE addresses is always an estimation based on observable output, not a direct observation of the layer itself. In SCOPE, self-consistency is estimated using multiple observed results as clues. This paper positions L3 as one such object of estimation.

As described above, L3 is not directly observed, but is positioned as a cognitive layer estimated from results observed as Malkuth. The next paper examines the possible functional correspondence of L3 between humans and LLMs.

Chapter 6: Correspondence with LLMs

Chapter 5 organized L3 as a cognitive layer estimated from results observed as Malkuth. This chapter examines the possibility of a functional correspondence between L3 in humans and LLMs.

In humans, L3 is positioned as the cognitive layer that maintains self-consistency within the individual. LLMs, on the other hand, are not considered to possess self-awareness or identity. For this reason, it cannot be assumed that the same kind of self-consistency exists in LLMs as it does in humans. In humans, self-consistency is thought to be maintained and to change through its relationship with experience and one's outlook on life. Whether a comparable structure exists in LLMs is not addressed in this paper.

In LLMs, however, behavior can be observed in which the model appears to preserve the overall consistency of its responses, or to maintain certain basic principles it is meant to uphold as a system. While this differs from self-consistency in humans, it may correspond structurally to a function that maintains consistency.

This paper positions this consistency-maintaining function in LLMs as a candidate structural function corresponding to L3 in humans, and refers to it as the Coherence Core (tentative term). This term is used to distinguish the consistency-maintaining function in LLMs from self-consistency in humans. It should be noted that this is a tentative term used to describe a functional correspondence, and does not imply that LLMs possess self-awareness or personality.

The correspondence with the Coherence Core discussed in this paper assumes systems such as large language models, which generate responses while retaining context. Systems that operate solely through simple state transitions are outside the scope of this paper.

Furthermore, this paper does not address the specific processing mechanism through which the Coherence Core is realized. This is left as a topic for future consideration, including its relationship to the internal structure of LLMs and to SCOPE as a whole.

As described above, this paper positions the possibility of a functional correspondence between L3 in humans and the Coherence Core in LLMs as a hypothesis.

Final Chapter

This paper has positioned L3 as the cognitive layer responsible for self-consistency, and has organized its basic characteristics, the possibility of change, its relationship with other layers, its observation through Malkuth, and its functional correspondence with LLMs.

L3's basic role is to maintain self-consistency. However, self-consistency is not fixed; it may change depending on the situation. In this paper, such change has been positioned not as a loss of self-consistency, but as something that occurs in order to preserve it.

At the same time, the mechanism by which self-consistency is maintained under certain conditions and changes under others cannot be fully explained by L3 alone. SCOPE presupposes a cognitive structure in which each layer mutually influences the others, and the maintenance and change of self-consistency must also be understood in relation to other layers.

The purpose of this paper has been to organize the role of L3 on its own. The structure underlying the maintenance and change of self-consistency will be examined separately within the SCOPE framework as a whole, in relation to the other layers.

The next paper addresses L4.